Lecture Scribe CSE400

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Generate a lecture scribe intended to serve as exam-oriented reference material for revision.

Use only the provided lecture slides and relevant textbook content as context. Do not introduce any material, explanations, examples, intuition, or assumptions that are not explicitly present in the provided context.

The scribe must faithfully reconstruct what was taught in the lecture and should allow a student to revise the material reliably for a closed-notes exam.

Include, exactly as presented in the lecture materials:

Definitions and notation

Assumptions and conditions

Statements of theorems or results

Proofs or derivations, step by step, when they appear in the lecture

Worked examples with intermediate steps, only if they appear in the lecture

Follow the structure and scope of the lecture strictly. Prioritize correctness and completeness. Do not invent or infer missing content.

Output the scribe as text in the chat itself.

Below is a **lecture scribe for CSE400 – Lecture 6**, reconstructed **strictly and only** from the provided lecture slides.
No material has been added, inferred, or extended beyond what explicitly appears in

the slides.
This scribe is intended to be **exam-oriented, closed-notes revision material** and follows the **exact scope, structure, definitions, results, and examples** presented in the lecture.

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

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1. Random Variables

Definition

A **random variable** X on a sample space Ω is a function

$$X : \Omega \to \mathrm{R}$$

that assigns to each sample point $\omega \in \Omega$ a real number.

Until further notice, attention is restricted to **discrete random variables**, i.e., random variables that take values in a **finite or countably infinite** set.

Although X maps Ω to R, the actual set of values

$$\{X(\omega) : \omega \in \Omega\}$$

is a **discrete subset** of R.

Distribution of a Random Variable

The distribution of a random variable can be visualized as a **bar diagram**:

- The x-axis represents the values the random variable can take.
- The height of the bar at value a is $\Pr[X = a]$.

- Each probability is computed from the probability of the corresponding event in the sample space.

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2. Guide to Selecting a Probability Distribution

Discrete Random Variable

- Countable support
- Probability **Mass** Function
- Probabilities assigned to **single values**
- Each possible value has **strictly positive probability**

Continuous Random Variable

- Uncountable support
- Probability **Density** Function
- Probabilities assigned to **intervals of values**
- Each possible value has **zero probability**

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3. Examples of Random Variables

Example 1: Tossing 3 Fair Coins

Let Y denote the number of heads that appear when 3 fair coins are tossed.

Possible values:

$$Y \in \{0, 1, 2, 3\}$$

Probabilities:

$$\Pr(Y = 0) = \Pr(t, t, t)$$

$$\Pr(Y = 1) = \Pr\{(t, t, h), (t, h, t), (h, t, t)\}$$

$$\Pr(Y = 2) = \Pr\{(h, h, t), (h, t, h), (t, h, h)\}$$

$$\Pr(Y = 3) = \Pr(h, h, h)$$

Since Y must take one of the values 0 through 3:

$$\sum_i \Pr(Y = i) = 1$$

4. Probability Mass Function (PMF)

Definition

A random variable that can take on at most a **countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range

$$R_X = \{x_1, x_2, x_3, \dots\}$$

(finite or countably infinite).

The function

$$p_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

Since X must take one of these values:

$$\sum_{k=1}^{\infty} p_X(x_k) = 1$$

PMF Example

Given:

$$p(i) = c\, e^{-i}, \quad i = 0, 1, 2, \dots$$

Since

$$\sum_{i=0}^{\infty} p(i) = 1$$

we have:

$$c \sum_{i=0}^{\infty} e^{-i} = 1$$

Using

$$\sum_{i=0}^{\infty} e^{-i} = \frac{1}{1 - e^{-1}}$$

implies:

$$c = e^{-1}$$

Results:

$$\Pr(X = 0) = e^{-1}$$

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2) = 1 - \Pr(X = 0) - \Pr(X = 1) - \Pr(X = 2)$$

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5. Bayes' Theorem (Recap)

Bayes Formula (Proposition 3.1)

$$\Pr(A \mid B_i) = \frac{\Pr(B_i \mid A)\Pr(A)}{\sum_{i=1}^{n} \Pr(A \mid B_i)\Pr(B_i)}$$

- $\Pr(B_i)$: **A priori probability**
- $\Pr(B_i \mid A)$: **Posteriori probability**

Example 1: Auditorium with 30 Rows

- 30 rows
- Row 1 has 11 seats, Row 30 has 40 seats
- A row is selected uniformly
- A seat within the selected row is selected uniformly

Tasks:

- Compute $\Pr(\text{Seat } 15 \mid \text{Row } 20)$
- Compute $\Pr(\text{Row } 20 \mid \text{Seat } 15)$

Example 2: Communication System (Receiver)

Conditional probabilities:

$$\Pr(0_r \mid 0_t) = 0.95, \quad \Pr(1_r \mid 0_t) = 0.05$$

$$\Pr(0_r \mid 1_t) = 0.10, \quad \Pr(1_r \mid 1_t) = 0.90$$

Tasks:

1. Find $\Pr(0 \text{ received})$ and $\Pr(1 \text{ received})$
2. Find posterior probabilities of transmitted bit given received bit
3. Find overall probability of error

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6. Independent Events

Definition

Two events A and B are **independent** if:

$$\Pr(A \mid B) = \Pr(A), \quad \Pr(B \mid A) = \Pr(B)$$

Hence:

$$\Pr(A, B) = \Pr(A)\Pr(B)$$

Mutual Independence (Three Events)

Events A, B, C are mutually independent if:

$$\Pr(A, B) = \Pr(A)\Pr(B)$$

$$\Pr(A, C) = \Pr(A)\Pr(C)$$

$$\Pr(B, C) = \Pr(B)\Pr(C)$$

$$\Pr(A, B, C) = \Pr(A)\Pr(B)\Pr(C)$$

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7. Types of Discrete Random Variables

7.1 Bernoulli Random Variable

Experiment outcome: **Success or Failure**

Define:

$$X = \begin{cases} 1 & \text{Success} \\ 0 & \text{Failure} \end{cases}$$

PMF:

$$p_X(0) = 1 - p, \quad p_X(1) = p, \quad p \in (0, 1)$$

7.2 Binomial Random Variable

Experiment:

- n independent trials
- Success probability p

Random variable:

$$X = \text{number of successes in } n \text{ trials}$$

Notation:

$$X \sim B(n, p)$$

PMF:

$$p(i) = \binom{n}{i} p^i (1-p)^{n-i}, \quad i = 0, 1, \dots, n$$

7.3 Geometric Random Variable

Experiment:

- Independent trials
- Success probability p
- Trials continue until first success

Random variable:

$$X = \text{number of trials until first success}$$

PMF:

$$\Pr(X = n) = (1-p)^{n-1} p, \quad n = 1, 2, \dots$$

Example: Urn Problem

- N white balls, M black balls
- Replacement after each draw

$$p = \frac{M}{M+N}$$

$$\Pr(X = n) = \left(\frac{N}{M+N}\right)^{n-1} \frac{M}{M+N}$$

7.4 Poisson Random Variable

Definition:
A random variable X taking values $0, 1, 2, \ldots$ is **Poisson** with parameter $\lambda > 0$ if:

$$\Pr(X = i) = \frac{e^{-\lambda}\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Normalization:

$$\sum_{i=0}^{\infty} \Pr(X = i) = 1$$

Note: Poisson RV approximates $B(n, p)$ when n is large, p is small, and np is moderate.

End of Lecture 6

If you want, I can next:

- Convert this into **condensed exam cheat notes**
- Extract **only definitions + formulas**
- Create **likely exam questions strictly from this lecture**
- Or format it as a **print-ready PDF revision sheet**